

CS120 Project Report

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Overview

Throughout this semester in the CS120 course, I successfully completed four projects. In Projects 1 and 2, I implemented Aethernet, a network that transmits signals based on sound. In Projects 3 and 4, I connected Aethernet to real-world networks, creating a complete network system. By completing this project, I gained a deeper understanding of the challenges faced in real-world network development and the techniques used to overcome them.

Project 0: Warm up

In Project 0, I used the Go language library ASIO to read and write audio signals. With ASIO, I was able to efficiently handle audio streams in Go. This project allowed me to become familiar with handling audio data, recording, and playback, laying the foundation for the subsequent audio transmission work.

Project 0 served as an introductory task to familiarize me with the fundamental tools and concepts used in audio-based networking. It focused primarily on the basics of audio processing, including capturing and playing back sound, and preparing for more advanced projects. The objective of this project was to understand the mechanics of sound data manipulation and processing within a programming environment (specifically Go). By the end of this project, I was able to record and playback audio, and also handle basic audio streams for further manipulation.

One of the main difficulties I encountered during this task was the configuration of the ASIO library with Go. The documentation for integrating ASIO with Go was sparse, so I had to dig into examples from the community and trial-and-error to get the system up and running. Another issue was related to ensuring that the audio was processed in real-time with minimal latency. Using ASIO's session model required careful handling of callbacks to manage input and output buffers efficiently.

Project 1. Acoustic Link

Project 1 focused on creating an acoustic link for transmitting data via sound waves. The goal was to generate sound waves, modulate the data into these sound waves, and then decode the received sound back into data. This task involved exploring various modulation techniques such as Phase Shift Keying and the challenges of error correction in an acoustic communication channel.

Task1 was simple. The only thing I need to do is to generate a simple audio signal. In Go, I used the ASIO library to handle audio input and output, setting the sample rate to 44100Hz, which is the standard audio sampling rate that covers most of the audio frequency range. I generate two sine wave signals, one at 1000Hz and one at 10000Hz, and mixes them to output through the audio interface. Next, I needed to convert the binary 01 stream into an audio signal for transmission. During this process, I designed an easily recognizable preamble to help the receiver correctly identify the start of the signal.

Throughout the transmission of the audio signal, I faced many unstable factors, particularly environmental noise that interfered with the signal. To minimize this interference, I chose to output the audio on the left channel only, which significantly improved the accuracy of the signal by avoiding interference from the right channel and other potential sources. Additionally, I used headphones pressed tightly to the audio output port to reduce the noise picked up by the computer's microphone, ensuring more accurate transmission.

During practical testing, due to the low sensitivity of the external microphone, the transmission accuracy would greatly decrease if I used it directly for reception. By pressing the headphones close to the output port, I effectively minimized this problem, allowing the audio signal to be transmitted more clearly and accurately.

The difficulty here was to ensure that the modulation process was robust enough to handle real-world noise, which could distort the signal and affect the accuracy of the data transmission. Initially, I encountered issues with low signal-to-noise ratios. To mitigate this, I used a preamble signal to mark the start of each transmission, which helped with synchronization. Additionally, noise from the computer's fan and microphone sensitivity issues impacted the clarity of the transmitted data, so I had to find ways to minimize these disturbances, such as using headphones tightly pressed against the output port to improve the reception.

Project 2. Manage Multiple Access

In Project 2, I improved the audio transmission method with the goal of transmitting more data in less time and reducing noise.

However, issues with amplitude setting, noise interference, and real-time processing capabilities resulted in some decoding errors. When handling the ACK packets, the latency further hindered real-time processing. To address this, I attempt to change the packet size to reach higher accuracy.

Project 3. To the Internet

In this project, I used virtual network interfaces to connect two computers.

In task0, I started by learning how to send and receive IP datagrams over the acoustic network. The challenge was in ensuring that the IP packets could be correctly encapsulated within the audio signals and transmitted without data loss.

And in the following task. First, virtual network interfaces were created on both computers, and IP addresses and gateways were set for each interface. Then, I connected them using an audio cable. Next, I configured the routing table on Node2 to route ping packets through the audio cable instead of through traditional network interfaces. Specifically, Node2 forwarded data from the virtual network interface to the physical layer, transmitting data through the audio cable. Node1, upon receiving the signal, captured the packets from the physical layer, then forwarded them to Node2 through the virtual network interface, completing the ping ACK response and achieving successful communication between the

two computers via the audio line.

Then I configured Node1 as a relay station. In this setup, Node3 first sent a ping packet to Node1, which then forwarded the ping packet to Node2. This process was similar to the previous audio forwarding, except that Node1 acted as the intermediary node in the forwarding process.

Next, I implemented a method similar to NAT to allow Node2 to access the internet through Node1. Specifically, Node1 acted as a forwarding node that forwarded Node2's requests to the external network. This approach, while not a traditional NAT, involved re-packaging the requests and forwarding them to the external network. Node1 also established a port mapping table to ensure that each data packet was correctly routed to its destination. This method enabled Node2 to communicate with the external network via the audio link.

Project 4: Above IP

Project 4's task was to allow Node2 to access DNS servers through Node1 and retrieve the IP address of a target website. To achieve this, I continued using the virtual network interface method from Project 3, utilizing the audio cable and virtual network interfaces to link Node2 and Node1.

First, Node2 sent a DNS request to Node1, which forwarded the request to an external DNS server. The DNS server resolved the target website's IP address and returned it to Node2. Afterward, Node1 used the resolved IP address to forward ping requests, achieving successful communication between Node2 and the external network.

In this project, I also faced challenges in configuring the routing table. To ensure that DNS requests were correctly forwarded, I had to configure the appropriate DNS and external IP mappings in Node1's routing table. Additionally, during packet capturing, I had to specifically mark the required packets to ensure that the system could properly identify and process the DNS requests and responses.

One of the main challenges was debugging the state transition logic between Node1 and Node2, especially when handling large data packets. Due to synchronization issues in the multithreaded framework, some data packets were not properly parsed. Despite these difficulties, I successfully implemented DNS resolution and was able to forward DNS requests to the external network through Node1.

These four projects presented many challenges, particularly in audio transmission, network connection, and protocol implementation. Through the use of virtual network interfaces and audio signal transmission, I successfully established communication between different nodes and, in Projects 3 and 4, connected Aethernet to real-world networks, forming a complete network system.

Through these projects, I gained hands-on experience in building real-world systems that handle both audio-based communication and traditional network protocols. The tasks helped me understand the complexities involved in both low-level signal processing and higher-level network communication. In particular, the challenges of handling real-time audio and maintaining reliable communication in noisy environments were valuable learning experiences that provided insights into practical network development.

Throughout the projects, I used the Go language ASIO library to efficiently process audio signals and utilized virtual network interfaces to connect two computers. Despite facing challenges such as real-time processing delays, routing table configuration, and audio interference, I successfully resolved many of these issues by adjusting the audio output and implementing methods similar to NAT.

Suggestion and feeling

The CS120 course has been a challenging yet rewarding experience, and it has provided valuable insights into the complexities of computer networks. However, there are areas where the course structure and project design could be improved to better support students in their learning journey. One of the main challenges I encountered during the projects was the debugging process. (especially in project 1,2) Running the code in real-life scenarios requires attention to a wide range of issues, from software bugs to hardware-related problems, all of which demand considerable time and effort to resolve. This often resulted in frustration, as the debugging phase consumed more time than anticipated, making it difficult to progress quickly. This difficulty might be alleviated by adjusting the difficulty curve of the course projects.

The current project arrangement is effective in giving students hands-on experience with network concepts and allowing them to explore classical solutions to real-world networking problems. However, it would be beneficial if the projects were structured in a way that gradually builds upon foundational concepts, starting with simpler tasks and advancing to more complex topics. This would help students become familiar with the fundamental ideas before tackling more advanced concepts. When faced with a new project, I always feel like I'm starting from scratch! Additionally, the course would benefit from breaking the projects into distinct phases, with each phase focusing on a specific networking technology or technique. This approach would allow students to gradually master each topic, ensuring a smoother learning process and greater retention of knowledge.

Another area for improvement is the guidance provided during the course projects. While the course content is rich and comprehensive, the lack of clear guidance often makes me spending significant time exploring secondary content. Providing more structured guidance or resources would help students navigate through the projects more efficiently, reducing the frustration of not knowing where to seek help. (sadly only project 0 has structured guidance.) This could be further addressed by introducing a mid-project evaluation system, where students' progress is assessed, and feedback is given. This would allow students to understand their performance and make necessary adjustments in real-time, improving the

overall learning experience.

Furthermore, I believe that introducing real-world case studies into the curriculum would significantly enhance the practical aspect of the course. Students could explore how specific networking tools and technologies are applied to solve real-world problems, or analyze case studies of advanced networking solutions. This would help bridge the gap between theory and practice, providing students with a clearer understanding of how the concepts they learn in class are used in industry settings.

Finally, regular feedback mechanisms—whether through face-to-face meetings, online discussions, or written comments—would offer students timely insights into their strengths and areas for improvement. This continuous feedback loop would help students stay on track throughout the course and address any confusion or difficulties they might encounter. With these improvements, the course would not only help students gain a deep understanding of theoretical concepts but also equip them with the practical skills necessary to succeed in the ever-evolving field of computer networks. By refining the project structure, providing more guidance, and fostering collaboration, the course could better meet students' needs and contribute to their overall development.